

# R&S Oscilloscope Automated Pressure Waveform Saves with Raspberry Pi

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Figure 1: The pressure saving GUI.

# 1 Introduction

The purpose of this project is to automate the saving of oscilloscope waveform data with the pressure measured by a pressure sensor attached to a vacuum chamber for levitated nanoparticles.

Before, data was saved manually on an oscilloscope by looking at a webcam and attempting to time the data save when the desired pressure was reached. This new method ensures consistency when saving this data.

## 2 Hardware Requirements

An overview of the hardware used, both for the optical setup and pressure saving side are shown below.

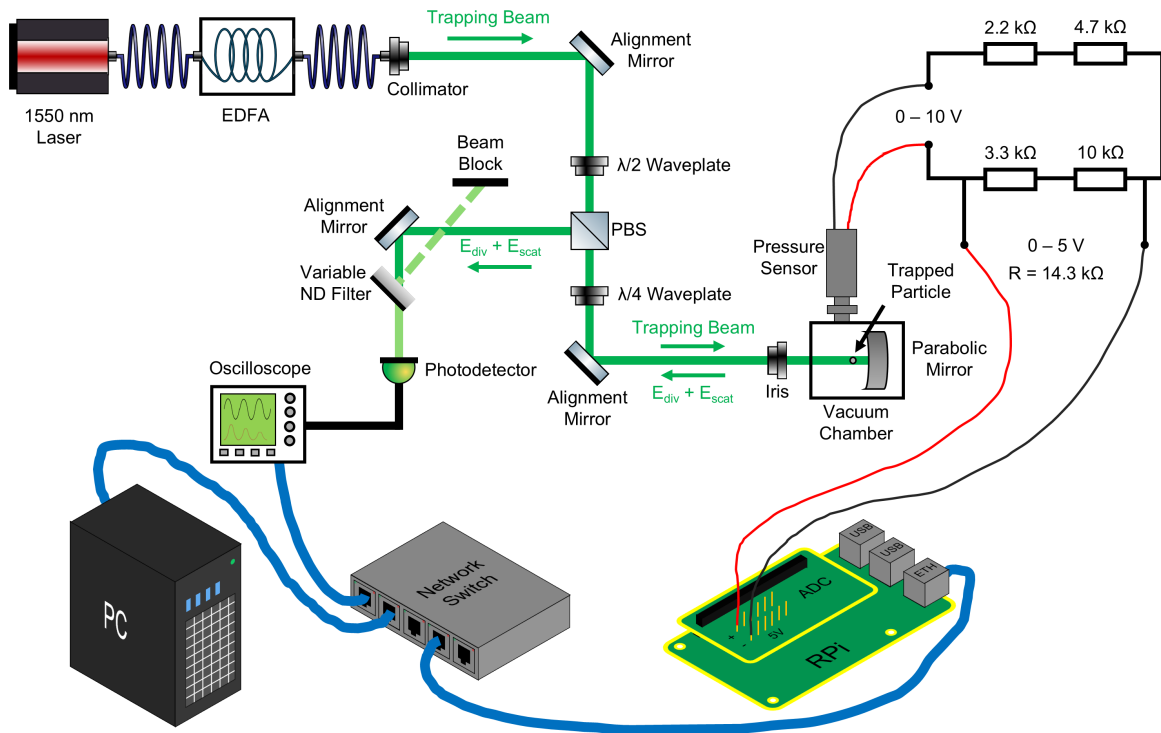


Figure 2: The hardware.

Focusing the the pressure saving hardware, it can be summarised as follows:

- Raspberry Pi (used here is a [Raspberry Pi 5](#))
- Micro SD card
- [Raspberry Pi ADC](#)
- Network switch with at least three ethernet ports
- Three ethernet cables
- Lab PC
- Resistors of values 2.2 kΩ, 4.7 kΩ, 3.3 kΩ and 10 kΩ
- Breadboard

- Male to male, male to female and female to female GPIO leads
- Solder and soldering iron
- [Pressure gauge](#)

## 3 Software Requirements

The software is split into the requirements for the Raspberry Pi and Lab PC.

### 3.1 Raspberry Pi

A version of Python is required here, as well as several packages. It is best to create a virtual environment for the packages to be installed to in order to avoid any conflicts for version numbers.

The following packages must be installed using the package manager `pip`

- `pip install customtkinter`
- `pip install numpy`
- `pip install matplotlib`
- `pip install RSInstrument`

Also, grab the Python file `ADCPi.py` from the GitHub page [here](#) and save it in the same directory as the files `GUI_RSpress19.py` and `RS_save_data.py` on the Raspberry Pi.

### 3.2 Lab PC

On the Lab PC side, the only software required is the remote access utility [MobaXterm](#). This enables SSH access to the Raspberry Pi and display of the graphical user interface over a network connection, something that other tools such as Putty lack. Make sure that the Raspberry Pi has X11 GUI output enabled. Also, note the IP address of the Raspberry Pi as this is required for network connection in MobaXterm.

## 4 Running the Code

Using the program MobaXterm on the Lab PC, SSH into the Raspberry Pi with a username and password. Then run the following commands:

```
1 cd Documents/NATS6007/ # change directory
2 source RSPress/bin/activate # activate the virtual environment RSPress
3 python3 GUI_RSpress19.py # run the current version of the code
```

Listing 1: Instructions to run the Python code